

Multi-asset benchmark — final results

Coinbase BTC / ETH / SOL, 7 days Jan 2026 — final_v2 protocol per coin

Final report 2026-07-14. 6 models $\times$ 3 training seeds $\times$ 3 rollout seeds per coin; seq/stride 1024, 12 epochs (equal step counts across models within each coin), val-split rate calibration with full-scale verification (SAHP uncalibrated by protocol, †), lossless streaming prediction, carried-state rollouts, equal-duration real-vs-sim comparison. **Status:** FINAL. All 54 arms complete; each coin passes its strict completeness check (REPORT_OK: every genuine json, every (seed $\times$ rollout) stylized-facts file for non-excluded arms, all required metrics finite). All calibration exclusions are confirmed reproducible (see §4).

1. Run status

coin	complete	train fail (infra)	SF fail (calibration)	running
btc	15	0	3	0
eth	16	0	2	0
sol	15	0	3	0

All infrastructure train failures (NFS stale-handle cache race, “CUDA device busy”) were rerun successfully; every checkpoint trained and completed prediction scoring. The only exclusions are *substantive*, reproducible calibration failures, all in the S2P2 family (§4).

2. Prediction (per genuine event; mean $\pm$ 95% CI over available seeds)

Bold = best per column: lower is better except ACC (higher) and mean_u (closest to 1).

BTC

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	-1.067$\pm$0.014	-3.215$\pm$0.004	2.148$\pm$0.016	0.227$\pm$0.006	1.07 $\pm$ 0.00	0.03$\pm$0.00	0.449$\pm$0.003	8.57$\pm$0.13
lstm	-0.437 $\pm$ 0.400	-2.710 $\pm$ 0.343	2.273 $\pm$ 0.063	0.305 $\pm$ 0.034	1.44 $\pm$ 0.54	0.03 $\pm$ 0.00	0.411 $\pm$ 0.015	9.71 $\pm$ 0.62
sahp	-0.928 $\pm$ 0.014	-3.189 $\pm$ 0.019	2.261 $\pm$ 0.006	0.256 $\pm$ 0.023	0.90 $\pm$ 0.06	0.03 $\pm$ 0.00	0.412 $\pm$ 0.001	9.59 $\pm$ 0.06
pct-lstm	-0.598 $\pm$ 0.079	-3.176 $\pm$ 0.048	2.579 $\pm$ 0.035	0.242 $\pm$ 0.042	1.01$\pm$0.11	0.03 $\pm$ 0.00	0.313 $\pm$ 0.010	13.18 $\pm$ 0.47
s2p2	-0.397 $\pm$ 0.235	-2.578 $\pm$ 0.245	2.181 $\pm$ 0.019	0.330 $\pm$ 0.037	2.36 $\pm$ 0.21	1.47 $\pm$ 2.69	0.439 $\pm$ 0.003	8.86 $\pm$ 0.17
ss2p2-full	-0.865 $\pm$ 0.109	-3.018 $\pm$ 0.116	2.152 $\pm$ 0.008	0.265 $\pm$ 0.001	1.85 $\pm$ 0.13	0.03 $\pm$ 0.00	0.447 $\pm$ 0.003	8.61 $\pm$ 0.07

ETH

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	-0.812 $\pm$ 0.033	-3.412 $\pm$ 0.014	2.600 $\pm$ 0.019	0.254 $\pm$ 0.012	1.03$\pm$0.05	0.03$\pm$0.00	0.341 $\pm$ 0.006	13.46 $\pm$ 0.26
lstm	0.154 $\pm$ 0.038	-2.606 $\pm$ 0.029	2.760 $\pm$ 0.009	0.336 $\pm$ 0.005	1.66 $\pm$ 0.01	0.03 $\pm$ 0.00	0.288 $\pm$ 0.003	15.80 $\pm$ 0.14
sahp	-0.598 $\pm$ 0.008	-3.321 $\pm$ 0.011	2.723 $\pm$ 0.005	0.278 $\pm$ 0.057	0.95 $\pm$ 0.21	0.04 $\pm$ 0.01	0.297 $\pm$ 0.004	15.23 $\pm$ 0.08
pct-lstm	-0.342 $\pm$ 0.103	-3.302 $\pm$ 0.077	2.960 $\pm$ 0.029	0.247 $\pm$ 0.069	1.12 $\pm$ 0.15	0.03 $\pm$ 0.00	0.248 $\pm$ 0.007	19.30 $\pm$ 0.55
s2p2	-0.381 $\pm$ 0.224	-3.011 $\pm$ 0.197	2.629 $\pm$ 0.027	0.309 $\pm$ 0.015	2.23 $\pm$ 0.26	10.16 $\pm$ 31.00	0.331 $\pm$ 0.008	13.87 $\pm$ 0.38
ss2p2-full	-0.866$\pm$0.084	-3.446$\pm$0.076	2.579$\pm$0.007	0.212$\pm$0.001	1.64 $\pm$ 0.08	0.03 $\pm$ 0.00	0.345$\pm$0.003	13.19$\pm$0.10

SOL

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	0.528 $\pm$ 0.014	-2.707 $\pm$ 0.004	3.235 $\pm$ 0.014	0.321 $\pm$ 0.008	1.15$\pm$0.06	0.05 $\pm$ 0.00	0.184 $\pm$ 0.004	25.40 $\pm$ 0.35
lstm	1.343 $\pm$ 0.250	-2.067 $\pm$ 0.218	3.410 $\pm$ 0.040	0.420 $\pm$ 0.025	1.56 $\pm$ 0.45	0.06 $\pm$ 0.00	0.141 $\pm$ 0.005	30.28 $\pm$ 1.22
sahp	0.668 $\pm$ 0.024	-2.687 $\pm$ 0.027	3.355 $\pm$ 0.019	0.376 $\pm$ 0.017	0.83 $\pm$ 0.04	0.06 $\pm$ 0.00	0.155 $\pm$ 0.003	28.65 $\pm$ 0.54
pct-lstm	0.895 $\pm$ 0.028	-2.607 $\pm$ 0.026	3.502 $\pm$ 0.009	0.314 $\pm$ 0.010	1.23 $\pm$ 0.02	0.06 $\pm$ 0.00	0.105 $\pm$ 0.004	33.18 $\pm$ 0.30
s2p2	0.671 $\pm$ 0.494	-2.560 $\pm$ 0.495	3.231 $\pm$ 0.001	0.204 $\pm$ 0.017	2.08 $\pm$ 0.52	46.18 $\pm$ 95.99	0.182 $\pm$ 0.002	25.30 $\pm$ 0.03
ss2p2-full	0.274$\pm$0.047	-2.911$\pm$0.044	3.185$\pm$0.003	0.173$\pm$0.007	1.57 $\pm$ 0.05	0.05$\pm$0.00	0.192$\pm$0.001	24.17$\pm$0.07

3. Stylized facts (rel-err vs real; rollout seeds averaged per checkpoint; mean $\pm$ 95% CI over checkpoints)

Checkpoints whose calibration failed verification produce no SF files and are excluded with a note; † = SAHP reported uncalibrated ($k=1$) by protocol.

BTC

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	37.92 $\pm$ 2.81	0.086 $\pm$ 0.080	0.863 $\pm$ 0.069	0.275$\pm$0.042	0.209$\pm$0.138	0.800 $\pm$ 0.090
lstm	38.51 $\pm$ 3.24	0.103 $\pm$ 0.093	0.993 $\pm$ 0.000	0.495 $\pm$ 0.404	0.537 $\pm$ 0.447	0.472 $\pm$ 0.060
sahp †	10.26 $\pm$ 13.17	0.706 $\pm$ 0.377	0.763$\pm$0.200	0.921 $\pm$ 2.546	0.745 $\pm$ 2.183	1.000 $\pm$ 0.000
pct-lstm	37.13 $\pm$ 1.90	0.063$\pm$0.054	0.786 $\pm$ 0.119	0.360 $\pm$ 0.288	0.333 $\pm$ 0.321	0.728 $\pm$ 0.092
s2p2 ($n=1$)	33.74	0.069	1.114	0.326	0.642	0.541
ss2p2-full ($n=2$)	43.29 $\pm$ 63.90	0.240 $\pm$ 1.830	0.788 $\pm$ 0.992	0.462 $\pm$ 0.618	0.384 $\pm$ 1.139	0.608 $\pm$ 0.167

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, ss2p2-full-s3

ETH

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	47.98 $\pm$ 7.02	0.373 $\pm$ 0.201	0.865 $\pm$ 0.110	0.518 $\pm$ 0.227	0.132$\pm$0.122	0.805 $\pm$ 0.000
lstm	48.78 $\pm$ 0.89	0.396 $\pm$ 0.025	0.994 $\pm$ 0.000	0.765 $\pm$ 0.294	0.497 $\pm$ 0.510	0.545 $\pm$ 0.000
sahp †	7.49 $\pm$ 3.76	0.786 $\pm$ 0.108	0.763 $\pm$ 0.456	0.435$\pm$0.927	0.468 $\pm$ 0.929	1.000 $\pm$ 0.000
pct-lstm	49.56 $\pm$ 9.14	0.418 $\pm$ 0.262	0.762$\pm$0.170	0.546 $\pm$ 0.504	0.367 $\pm$ 0.472	0.762 $\pm$ 0.174
s2p2 ($n=1$)	38.92	0.267	6.045	0.595	0.454	1.102
ss2p2-full	49.67 $\pm$ 5.90	0.421 $\pm$ 0.169	0.786 $\pm$ 0.046	0.810 $\pm$ 0.114	0.667 $\pm$ 0.018	0.671 $\pm$ 0.192

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2

SOL

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	23.93 $\pm$ 0.69	0.192$\pm$0.034	0.751 $\pm$ 0.167	0.721 $\pm$ 0.109	0.806 $\pm$ 0.071	0.799 $\pm$ 0.025
lstm	24.66 $\pm$ 0.75	0.228 $\pm$ 0.037	0.988 $\pm$ 0.000	0.969 $\pm$ 0.032	0.861 $\pm$ 0.100	0.500 $\pm$ 0.000
sahp †	6.40 $\pm$ 3.02	0.681 $\pm$ 0.150	0.925 $\pm$ 0.078	0.656$\pm$0.241	0.859 $\pm$ 0.063	1.000 $\pm$ 0.000
pct-lstm	24.58 $\pm$ 4.80	0.224 $\pm$ 0.239	0.460$\pm$0.335	0.715 $\pm$ 1.356	1.665 $\pm$ 2.536	0.626 $\pm$ 0.180
s2p2			<i>no calibrated checkpoints</i>			
ss2p2-full	26.08 $\pm$ 3.37	0.298 $\pm$ 0.168	0.598 $\pm$ 0.599	0.683 $\pm$ 0.624	0.778$\pm$0.156	0.591 $\pm$ 0.080

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, s2p2-s3

4. Calibration findings

s2p2 near-criticality worsens with event rate. 7 of 9 s2p2 checkpoints failed val-side calibration with full-scale verification (btc s1,s2; eth s1,s2; sol s1,s2,s3). The signature is identical everywhere: a near-discontinuous closed-loop rate response in the rescaling constant k (e.g. btc-s1: k 0.5930 $\rightarrow$ 0.5938 flips the probe rate 36 $\rightarrow$ 48.5 ev/s) with full-scale verify misses of 30–143%. Every retry reproduced its failure with a *bit-identical* bisection bracket (probes are seeded deterministically): these are properties of the checkpoints, not sampling noise. On SOL all three seeds failed — **s2p2 has no calibrated simulation arm on SOL**, a model-level exclusion, as with SAHP. The surviving s2p2 arms are btc-s3 and eth-s3 ($k=1.102$, verified). **ss2p2-full lost exactly one checkpoint of nine** (btc-s3, probe bisection non-convergence, also reproduced); the other eight calibrated and verified, including all three ETH seeds.

Reading the tables. Validation rates are btc 38.3, eth 47.9, sol 24.0 ev/s, so the fixed 1024-event context spans only $\sim$ 20–40s of market time — long-memory facts (clus/retACF) lean harder on carried state and TBPTT. Numbers are comparable *within* a coin; cross-coin comparisons should use relative errors only. CIs over fewer than 3 checkpoints are marked with their n ; $n=2$ uses $t_{0.975}=12.7$ and is wide by construction.